

Topological Parametrization and a Tentative Dynamical Action for the Swirl-String

A compact SST note in an orthodox string-theory order

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1 Introduction

This note is organized in a sequence intended to parallel the canonical development of original string theory: first the fundamental input, then the geometric parametrization of the string configuration, then the derived length scale, and finally a tentative action-like functional that selects dynamically preferred configurations.

In Swirl-String Theory (SST), a particle is modeled not as a point object but as a structured closed swirl-string in a three-dimensional incompressible, inviscid medium. Its mass-equivalent energy is assumed, at first approximation, to be controlled by the geometry and topology of the corresponding closed curve or knot configuration.

2 Fundamental input

We do not take r_c as primary input. Instead, the fundamental triplet is

$$(\omega_c, \rho_f, \mathbf{v}_\odot),$$

where:

- ω_c is the canonical Compton angular frequency,
- ρ_f is the effective fluid density,
- $\mathbf{v}_\odot$ is the characteristic swirl speed.

For the present canonical SST scale we take

$$\omega_c = 7.763440655383073 \times 10^{20} \text{ s}^{-1}, \quad \rho_f = 7.0 \times 10^{-7} \text{ kg m}^{-3}, \quad \mathbf{v}_\odot = 1.09384563 \times 10^6 \text{ m s}^{-1}.$$

In this parametrization, the characteristic microscopic length is not fundamental but derived.

3 Geometric parametrization of the swirl-string

A closed swirl-string is described by an embedding

$$\mathbf{X}(t, \sigma) = (x(t, \sigma), y(t, \sigma), z(t, \sigma)), \quad \sigma \in [0, 2\pi),$$

where t is the preferred medium time and σ is the internal string parameter.

A natural first class of explicit static configurations is given by (p, q) torus knots. Their centerline is parametrized by

$$x(\sigma) = (R + r \cos(q\sigma)) \cos(p\sigma), \quad (1)$$

$$y(\sigma) = (R + r \cos(q\sigma)) \sin(p\sigma), \quad (2)$$

$$z(\sigma) = -r \sin(q\sigma). \quad (3)$$

Here p counts poloidal windings and q counts toroidal windings. For $\gcd(p, q) = 1$ one obtains a single closed knot; for $\gcd(p, q) > 1$ one obtains a multi-component link.

This step plays the same formal role as the choice of embedding fields $X^\mu(\tau, \sigma)$ in orthodox string theory: one first specifies the configuration space of admissible string shapes, and only afterwards introduces the dynamical or effective selection principle.

4 Derived length scale

The characteristic core radius is derived from the fundamental input via

$$r_c = \frac{\mathbf{v}_\odot}{\omega_c}. \quad (4)$$

Numerically this gives

$$r_c = 1.40897017 \times 10^{-15} \text{ m}. \quad (5)$$

The torus scales are then written as

$$R = \lambda_R \frac{\mathbf{v}_\odot}{\omega_c}, \quad r = \lambda_r \frac{\mathbf{v}_\odot}{\omega_c}, \quad (6)$$

where λ_R and λ_r are dimensionless shape parameters.

A minimal tube cross-section is therefore

$$A_{\text{core}} = \pi r_c^2 = \pi \left(\frac{\mathbf{v}_\odot}{\omega_c} \right)^2, \quad (7)$$

which numerically yields

$$A_{\text{core}} = 6.23668012 \times 10^{-30} \text{ m}^2. \quad (8)$$

A corresponding effective line density is

$$\mu_{\text{eff}} \sim \rho_f A_{\text{core}}, \quad (9)$$

so that

$$\mu_{\text{eff}} \sim 4.36567609 \times 10^{-36} \text{ kg m}^{-1}. \quad (10)$$

5 A tentative Nambu–Goto-like SST action

To mimic the role played by the Nambu–Goto action in orthodox string theory, one may propose a preferred-time dynamical functional for the swirl-string centerline:

$$S_{\text{SST}}[\mathbf{X}] = \int dt \oint d\sigma \left[\frac{\mu_{\text{eff}}}{2} |\partial_t \mathbf{X}|^2 - \frac{T_{\text{eff}}}{2} |\partial_\sigma \mathbf{X}|^2 - \frac{B_\kappa}{2} \kappa^2 - \frac{B_\tau}{2} (\tau_g - \tau_0)^2 \right] - S_{\text{BS}}[\mathbf{X}]. \quad (11)$$

Here:

- T_{eff} is an effective line-tension coefficient,
- κ is the Frenet curvature of the centerline,
- τ_g is the geometric torsion,
- τ_0 is a preferred torsional bias for a chiral sector,
- S_{BS} is a nonlocal Biot–Savart self-interaction term.

A simple regularized form for the nonlocal term is

$$S_{\text{BS}}[\mathbf{X}] = \frac{\Lambda}{2} \int dt \oint d\sigma \oint d\sigma' \frac{\partial_\sigma \mathbf{X}(t, \sigma) \cdot \partial_{\sigma'} \mathbf{X}(t, \sigma')}{\sqrt{|\mathbf{X}(t, \sigma) - \mathbf{X}(t, \sigma')|^2 + \epsilon^2}}. \quad (12)$$

This is only a tentative SST analogue. Unlike the relativistic Nambu–Goto string, the medium selects a preferred time variable t , so Lorentz symmetry on the worldsheet and Weyl symmetry are not assumed here.

6 Static reduction and the effective knot functional

For time-independent configurations, the tentative action reduces to a static energy functional,

$$E_{\text{stat}}[\mathbf{X}] = \oint d\sigma \left[\frac{T_{\text{eff}}}{2} |\partial_\sigma \mathbf{X}|^2 + \frac{B_\kappa}{2} \kappa^2 + \frac{B_\tau}{2} (\tau_g - \tau_0)^2 \right] + E_{\text{BS}}[\mathbf{X}]. \quad (13)$$

At the knot-class level, this motivates the canonical SST effective mass functional

$$\mathcal{E}_{\text{eff}}[K] = \alpha C(K) + \beta L(K) + \gamma \mathcal{H}(K), \quad (14)$$

with the identifications:

- $\beta L(K)$ as the line-tension or length contribution,
- $\alpha C(K)$ as the near-contact or Biot–Savart contribution,
- $\gamma \mathcal{H}(K)$ as a helicity or Hopf-like stability term.

For a torus-knot family K_k representing successive stable configurations, the SST canon then uses the discrete-scale ansatz

$$\mathcal{E}_{\text{eff}}[K_k] \approx (\alpha C(K_0) + \beta L(K_0)) \varphi^{-2k}, \quad (15)$$

where $\varphi = \frac{1+\sqrt{5}}{2}$ is the golden ratio and k is the discrete generation index.

7 A tentative Polyakov-like rewriting

One may also attempt a Polyakov-like rewriting by introducing an auxiliary worldsheet metric h_{ab} on coordinates $\xi^a = (t, \sigma)$:

$$S_{\text{aux}}[\mathbf{X}, h] = -\frac{\lambda_0}{2} \int d^2\xi \sqrt{-h} h^{ab} \partial_a \mathbf{X} \cdot \partial_b \mathbf{X} - \int d^2\xi \sqrt{-h} U_{\text{loc}}[\kappa, \tau_g] - S_{\text{BS}}[\mathbf{X}]. \quad (16)$$

This rewriting is formally useful, but it should not be over-interpreted. In SST the ambient medium defines a preferred foliation, so the auxiliary metric is a calculational device, not evidence for an exact Weyl-invariant worldsheet theory.

8 Secondary constitutive relations

If one also introduces a core density ρ_{core} , the coarse-graining relation can be written as a secondary constitutive law:

$$\rho_f = K \Omega, \quad K = \frac{\rho_{\text{core}}}{\omega_c}. \quad (17)$$

Hence

$$\rho_f = \frac{\rho_{\text{core}}}{\omega_c} \Omega. \quad (18)$$

The important logical point is that this relation does not define the primary parametrization. The primary input remains

$$(\omega_c, \rho_f, \mathbf{v}_{\odot}),$$

while r_c , A_{core} , and other geometric scales are derived quantities.

9 Interpretation and status

Within this construction, particle families are not primarily distinguished by point-particle oscillator levels, but by topologically distinct closed swirl configurations. Torus knots provide a minimal explicit model class for:

- chiral sectors,
- generation-like hierarchies,
- and energy minima of the effective knot functional.

The present note should therefore be read as an SST analogue of the early kinematical and action-building stage of original string theory. The geometric embedding is explicit, the derived scale structure is explicit, and the proposed action links dynamically to the canonical knot-mass functional. What remains open is the full derivation of the coefficients T_{eff} , B_{κ} , B_{τ} , and Λ from first-principles SST fluid dynamics.

References

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